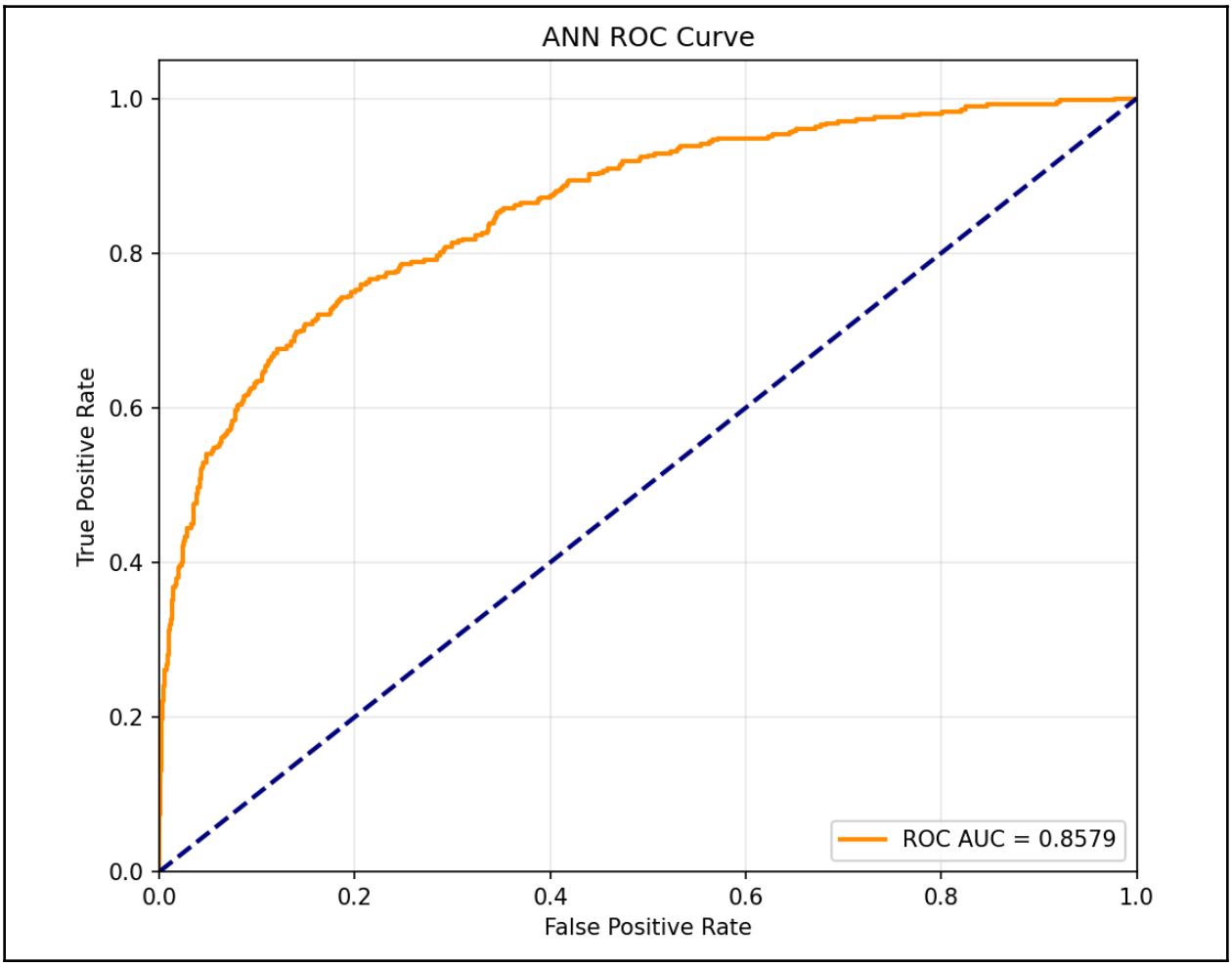
Experiment 7

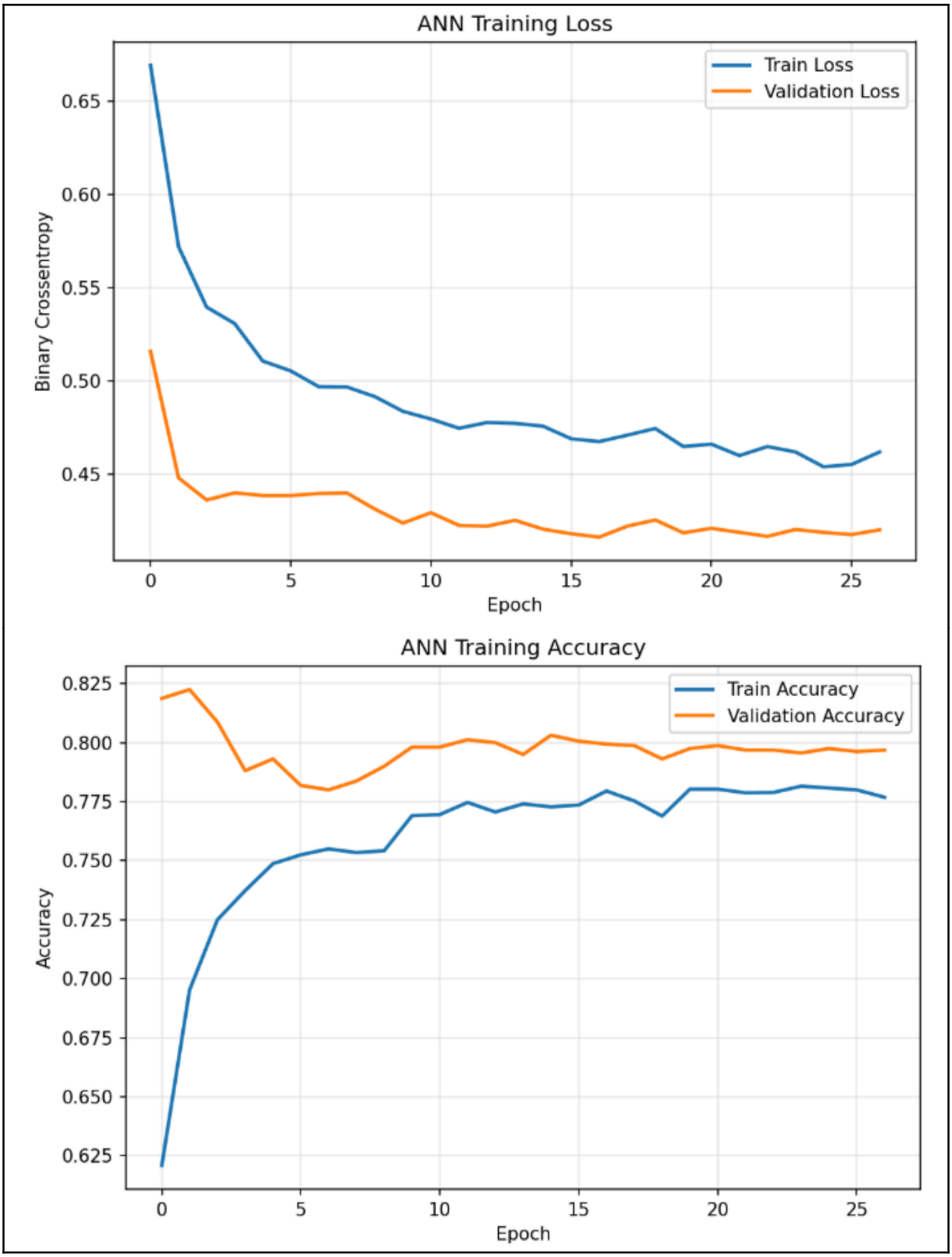
Aim : Build an Artificial Neural Network (ANN) using Keras/TensorFlow

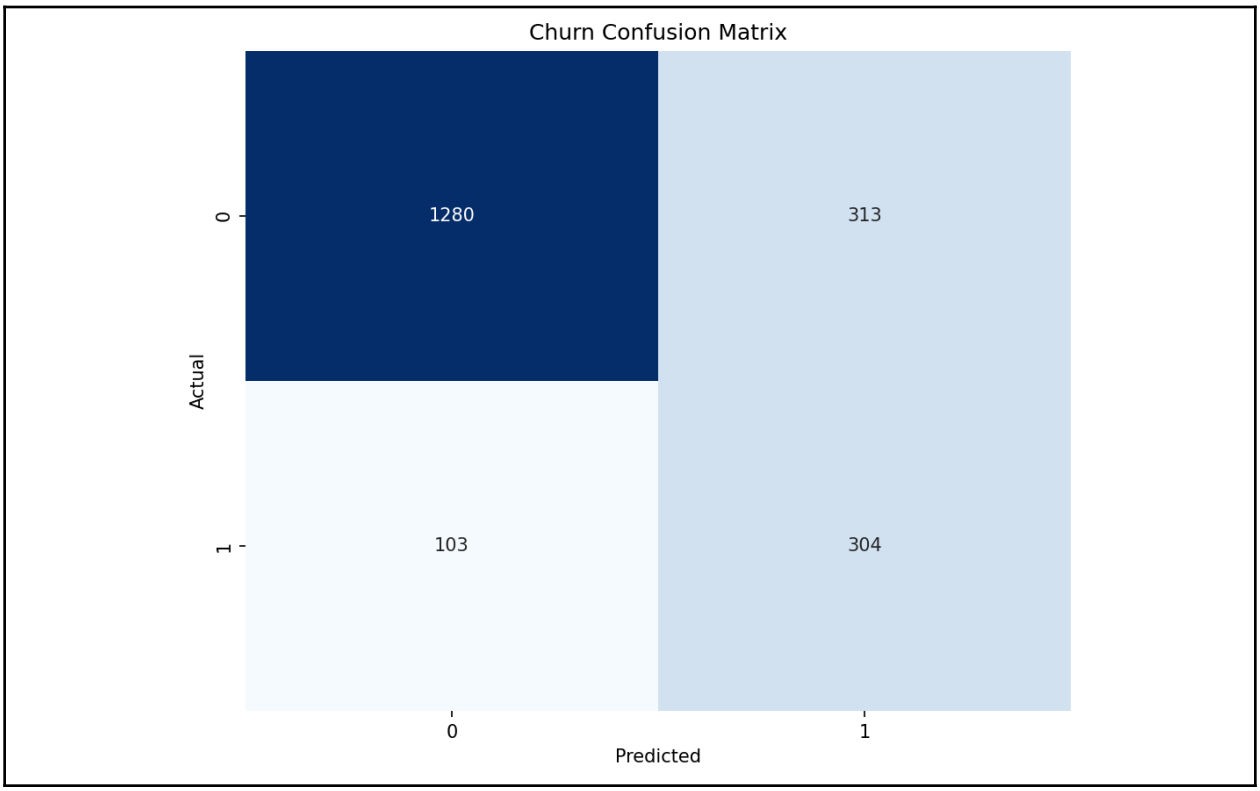
Data Source : [Churn Modelling | Kaggle]

- Contains bank customer data used to predict churn (whether a customer leaves the bank).
- It is a binary classification dataset (Exited: 0 = No, 1 = Yes).
- Around 10,000 customer records (commonly used version)
- Total 14 columns/features

Output :




ANN Training Loss
Train Loss
Validation Loss
Binary Crossentropy
0.65
0.60
0.55
0.50
0.45
0
5
10
15
20
25
Epoch
ANN Training Accuracy
Train Accuracy
Validation Accuracy
Accuracy
0.825
0.800
0.775
0.750
0.725
0.700
0.675
0.650
0.625
0
5
10
15
20
25
Epoch



Conclusion :

Building an ANN using Keras/TensorFlow provides a powerful and efficient way to implement deep learning models. Keras simplifies the process, making it accessible for beginners while leveraging the computational strength of TensorFlow.